



REVA Academy for Corporate Excellence

Program: MSc/M.Tech in Artificial Intelligence	Course: Computer Vision and Generative AI
Batch: AI 11	Exam: Final
Date: 27 th June 2026	Time: 2:00pm to 5:00pm
Duration: 3 Hrs	Marks: 100

Instructions	
1.	This is a closed book exam. No internet access will be provided. The exam must be written on lab systems only - personal laptops are not allowed. Any evidence of copying verbatim from any third-party resources or plagiarized contents will lead to suspension and other stringent actions. If any identical answers are found, both will be awarded zero.
2.	You may use any tool or software to answer the questions. All the outputs and codebase must be well commented.
3.	All the answers must be written in the answer script template provided by RACE. The template will be available in the LMS with the answer submission link. Along with the main answer script, all other artifacts like screenshots, codebase and any other materials must be submitted. All such artefacts must be named with the corresponding question number.
4.	All the artefacts must be zipped and submitted through the LMS Link. Zipped Folder must be named as 'SRN_Name of the Course'.

Section A				
Instructions		All the questions are mandatory. Answers should be brief and to the point.		Marks
Q1.	A manufacturing company wants to automatically inspect metal surface images to identify scratches, dents, and surface irregularities. Before training a deep learning model, the team wants to understand how classical image processing techniques can help highlight useful visual patterns.			
	a.	Explain how convolution works in image processing. Use the example of a simple filter/kernel to explain how pixel values are transformed.	2	6
	b.	Compare smoothing filters and edge detection filters. Why both might be useful in an industrial inspection pipeline?	2	

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	c.	In what situations would classical image processing still be useful even when deep learning models are available?	2	
Q2.	An object detection model is deployed at a construction site to detect whether workers are wearing safety helmets. During testing, the model sometimes misses helmets that are partially hidden and sometimes detects helmets where none are present.			
	a.	Explain true positive, false positive, false negative, and true negative in the context of this helmet detection system.	3	8
	b.	Why is Intersection over Union, IoU, important in object detection evaluation? Explain how IoU helps decide whether a predicted bounding box is correct.	3	
	c.	In this safety-critical use case, would you prioritize precision or recall? Justify your answer.	2	
Q3.	A radiologist wants to segment abnormal regions in MRI scans without manually drawing the full boundary of every region. The hospital is exploring whether a prompt-based segmentation model such as Segment Anything Model, SAM, can support this workflow.			
	a.	What is prompt-based segmentation? Explain how a user can guide a model such as SAM using prompts such as points, bounding boxes, or masks.	3	8
	b.	Explain the main components of SAM and how they work together as pipeline?	3	
	c.	Mention one advantage and one limitation of using SAM directly in domain-specific areas such as medical imaging or industrial inspection.	2	
Q4.	A disaster response team is analyzing aerial and satellite images after a flood. They want an AI system that can locate different objects or regions such as damaged houses, blocked roads, waterlogged areas, and rescue boats using simple text instructions, even if some of these object categories were not part of a fixed training label set.			

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	a.	What is meant by open-vocabulary object localization or “Locate Anything” in computer vision? How is it different from traditional object detection with fixed classes?	4	
	b.	Explain the role of Parallel Box Decoding in Locate Anything. How does it help the model locate multiple objects or regions efficiently?	3	9
	c.	Mention one advantage and one challenge of using Locate Anything-style models in real-world applications such as remote sensing, disaster management, or industrial inspection.	2	
A startup wants to use Vision Transformers for classifying satellite images, but the engineering team is more familiar with CNN-based models.				
Q5.	a.	Explain how a Vision Transformer converts an image into patches and processes them as tokens.	4	
	b.	How is the idea of attention useful in Vision Transformers for understanding relationships between different parts of an image?	3	9
	c.	Mention one advantage and one limitation of Vision Transformers compared with CNNs.	2	
A design team wants to generate synthetic product images for concept visualization. At the same time, an industrial inspection team wants to generate rare defect images to improve model training.				
Q6.	a.	Explain the basic idea of image generation using diffusion models.	4	
	b.	How are diffusion models different from GANs at a high level?	4	10
	c.	Mention one useful application and one risk or limitation of synthetic image generation in computer vision	2	

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Section B	
Instructions	All the questions are mandatory. 1. Use Google Colab to train the model with GPU support for faster execution. 2. Limit the number of EPOCHS to 10 or 20 to ensure the entire process runs within 10-15 minutes. 3. Accuracy of the model is not the primary concern; focus on the overall process, workflow, and understanding of the problem. 4. Use proper markdowns to explain steps and add comments in your code to ensure clarity and easy understanding for others. 5. PyTorch is recommended for implementation, but using other deep learning frameworks like TensorFlow or Keras is also acceptable. 6. Submit the notebook, output screenshots, trained model files if applicable, and any other supporting artifacts. 7. All artifacts must be named with the corresponding question number.

Q7.	You are working as an AI/ML engineer for a smart city analytics team. The team wants to build an image classification model that can classify outdoor scene images into different categories such as buildings, forest, glacier, mountain, sea, and street. Such a model can be useful for organizing large-scale image collections and supporting scene understanding applications. Use the Intel Image Classification Dataset. Dataset Link: https://www.kaggle.com/datasets/puneet6060/intel-image-classification The dataset contains natural scene images belonging to multiple classes.
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	a.	Data Preparation: Download and prepare the dataset for training and testing. Your tasks: • Load the image dataset. • Understand the class labels and folder structure. • Split the dataset into training and testing sets if a split is not already provided. • Apply standard preprocessing steps such as resizing and normalization. Apply suitable data augmentation techniques such as horizontal flipping, random rotation, color jitter, or random crop.	5	
	b.	Model Training using Transfer Learning: Train a multiclass image classification model using transfer learning. Your tasks: • Select a suitable pretrained CNN model such as ResNet any equivalent model. • Replace or modify the final classification layer according to the number of classes in the dataset. • Freeze or fine-tune the pretrained layers based on your chosen strategy. • Train the model for a limited number of epochs. Display training and validation accuracy/loss during training.	10	25
	c.	Model Evaluation and Inference: Evaluate the trained model and demonstrate inference on new images. Your tasks: • Plot training and validation accuracy/loss curves.	10	

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		- Evaluate the model on the test dataset. - Display the confusion matrix or classification report. - Explain how overfitting can be detected from the training curves. - Write a function to save the trained model. Write a function to load the saved model and predict the class of a new image.		
		You are working for a manufacturing quality inspection team. The team wants to automatically detect surface defects in steel or metal surface images. The goal is to train an object detection model that can locate visible defects using bounding boxes. Use the NEU Surface Defect Dataset from Roboflow Universe. Dataset Link: https://universe.roboflow.com/pavithraa-sekar/neu-surface-defect-dataset Download the dataset in YOLO format suitable for YOLOv8 or a similar object detection framework.		
Q8.				
	a.	Data Preparation: Prepare the object detection dataset for training. Your tasks: - Download the dataset in YOLO format. - Verify that the dataset contains training, validation, and test folders. - Check that the annotation files are available in YOLO format. - Verify that the data.yaml file correctly contains the dataset paths and class names. Display a few sample images with bounding box annotations	5	25

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	b.	Model Training and Evaluation: Train a YOLO-based object detection model for detecting surface defects. Your tasks: • Use YOLOv8 nano, YOLOv8 small, or any equivalent lightweight YOLO model. • Train the model for 10 to 25 epochs. • Display training loss and validation metrics. • Report important object detection metrics such as precision, recall, mAP@0.5, and mAP@0.5:0.95. Briefly explain what these metrics indicate about the model performance.	10	
	c.	Model Inference: Run inference using the trained object detection model. Your tasks: • Load the trained YOLO model. • Run inference on test images or new sample images. • Display the detected defects with bounding boxes and class labels. • Save at least three inference output images. Briefly explain possible reasons for false positives and false negatives in surface defect detection.	10	
